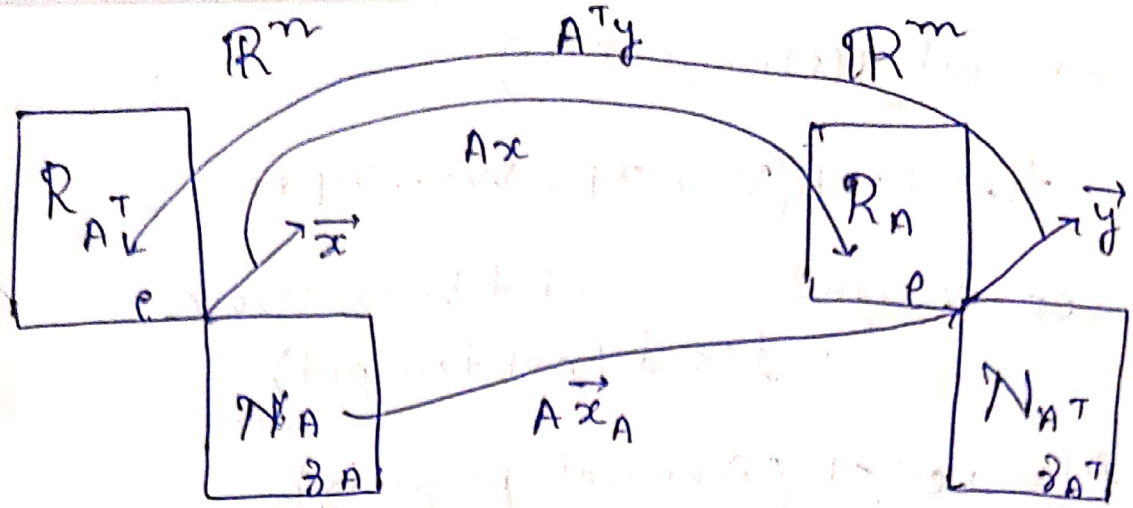




$$A = \sum_{j=1}^p s_j u_j u_j^T$$

Rank 1 Decomposition of A
Low Rank appear of A

$$A A^T \in A A^T$$

Eigen values of $A^T A \in A A^T$
are the same

Orthogonal Eigen vectors
of $A^T A$ form the ONB.

for $\text{colspace}(A^T)$.

$A \in A^T$ have same rank

If A is a full rank matrix ~~for rank~~
 $m \times n$ matrix ~~minimum~~
and A^T will not be a full rank matrix
if $m \neq n$.

$\Rightarrow A^T A$ will be a real symmetric matrix (RSM
of rank $n \Rightarrow$ Eigen values.

If the above happens, $\Rightarrow$ Invertible matrix

$\Rightarrow \det(A^T A) \neq 0 \Rightarrow$ Hence ~~each~~ none of the Eigen values are zero.

$AA^T \Rightarrow$ matrix of $m \times m$ dimension

Real symmetric Matrix

It should have m Eigen values. (Full Rank matrix)

But, we know that

$$\boxed{\text{Rank}(A) = \text{Rank}(A^T) = \text{Rank}(A^T A) = \text{Rank}(A A^T)}$$

id, since $\text{Rank}(A A^T) \leq m$

$\Rightarrow n$ non-zero eigen values and

$m-n$ eigen ~~vectors~~ values are zero.

Tall matrix (since, $m \geq n$).

~~I have~~ I have sum form of SVD as

$$A = \sum_{j=1}^p s_j v_j u_j^T$$

Now, product form of SVD is

$$A = U \Sigma V^T \text{ mit } m \times n \text{ für } \Sigma \text{ und } U = U^T U$$

Can I go from Product form to Sum Form
(or) vice versa? or not possible?

$$A = U \Sigma V^T$$

$$\text{Rank}(A^T A) = \text{Rank}(A^T A)^T = \text{Rank}(A^T) = \text{Rank}(A)$$

$$\begin{bmatrix} u_1 & u_2 & u_3 & \dots & u_n & \begin{matrix} \text{and} \\ \text{has regular eigenspaces} \end{matrix} \\ \chi_1^T & \chi_2^T & \chi_3^T & \dots & \chi_m^T \end{bmatrix}$$

$$m \times n \quad \left\{ \begin{array}{c|c} \begin{matrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_e \\ 0 & 0 & \dots & 0 \end{matrix} & \begin{matrix} 1 & 0 & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \\ 0 & 0 & \dots & 0 \end{matrix} \end{array} \right\} \quad n \times n$$

simplify, we get

$$[u_1 u_2 \dots u_e] \begin{bmatrix} \chi_1 & \chi_2 & \dots & \chi_{m-e} \end{bmatrix}^{m \times m}$$

$$\begin{bmatrix} s_1 v_1^T \\ s_2 v_2^T \\ \vdots \\ s_e v_e^T \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^{m \times n}$$

$$= s_1 u_1 v_1^T + s_2 u_2 v_2^T + s_3 u_3 v_3^T + \dots + s_e u_e v_e^T$$

Notations have been changed, that's it

Given $A^{m \times n}$, $x \in \mathbb{R}^n$

What does A do to x to result in b ?

$$A = \sum_{j=1}^e s_j v_j u_j^T$$

$$A x = \sum_{j=1}^e s_j v_j \underbrace{u_j^T x}_{u_j^T x \Rightarrow 1}$$

That means A operates on x
 will happen, by projecting x

on Orthormal Basis for Colspace(A^T)

And, then changes into ~~m~~ ~~direction~~
 dimensional axes

Similarly, $A^T = \sum_{j=1}^p \frac{1}{s_j} u_j v_j^T$

Hence, $\vec{x} = \sum_{j=1}^p \frac{1}{s_j} u_j v_j^T b$

Takes $\vec{b}$ and first of all projects the $\vec{b}$

to the Colspace(A) or $R(A)$

$\Rightarrow$ Because $\vec{b}$ should exist in Colspace(A)

Then, solution exists

$\Rightarrow$ either unique / Infinitely many

So, $\vec{x}$ will be obtained

$\Rightarrow$ But many $\vec{x}$ possible

$\Rightarrow$ I should completely project into Colspace
to get optimal solution

Optimal solution, in the sense that
makes least Square Error minimal

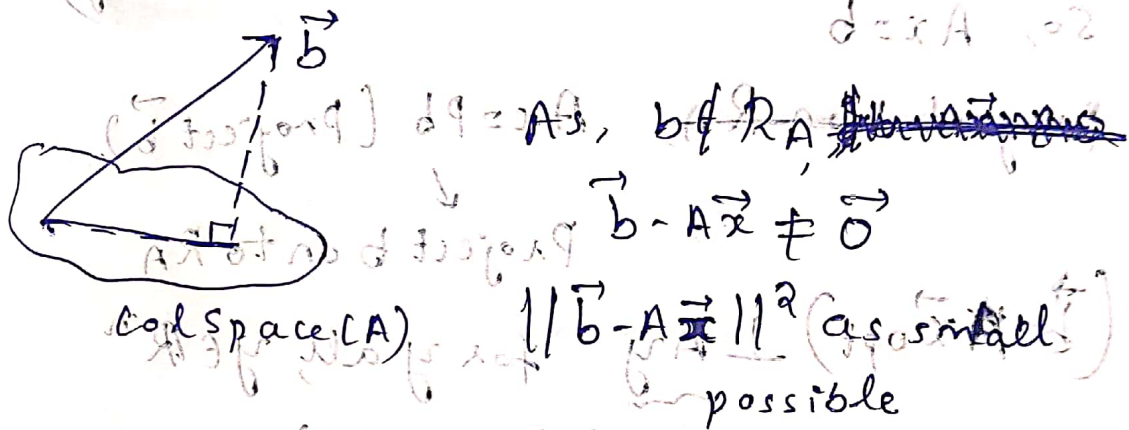
$\vec{x} = \vec{x}_P + \alpha \vec{x}_H$. For $\vec{x}$ to be minimal

$\alpha = 0 \Rightarrow$ Hence

$\vec{x}$ shouldn't have any component in N_A .

But still doubt? Many doubts?

we get $A\vec{x} = \vec{b}$ where $\vec{b} \in \text{Colspace}(A)$
(if no solution exists.)



square of this magnitude $\Rightarrow$ variance

talks about the distribution $(\vec{x}_A - \vec{b})$

$\Rightarrow$ Minimizing the variance (distribution)
 $\Rightarrow$ Points come closer

Approximation (not to the solution)

but, ~~the~~ to arrive at the solution

Do not distance (error) to the solution

And, of course, the longest distance is the one that is the furthest from the solution.

$A\vec{x} = \vec{b}$ $A^{n \times n}$, a full Rank matrix
rank(A) = n

$$\|\vec{b} - A\vec{x}_{opt}\|^2 \leq \|\vec{b} - A\vec{x}\|^2$$

(It's like a Bull Eye)

LSE (Least Square Error) can give

Optimal solution (Unique/Infinitely many)

So, $A\vec{x} = \vec{b}$

~~Project $\vec{b}$ on RA~~ $A\vec{x} = P\vec{b}$ (Project $\vec{b}$)

$\vec{0} \neq \vec{b} - A\vec{x}$

Project $\vec{b}$ on to RA

$(\vec{b} - A\vec{x}_{opt}) \perp Ay$ for all $y \in \mathbb{R}^n$

Colspace(A)

$\Rightarrow (\vec{b} - A\vec{x}_{opt}) \cdot (Ay) = 0$ for all $y \in \mathbb{R}^n$

$(Ay)^T (\vec{b} - A\vec{x}_{opt}) = 0$ for all $y \in \mathbb{R}^n$

$y^T A^T (\vec{b} - A\vec{x}_{opt}) = 0$

$\Rightarrow y^T (A^T \vec{b} - A^T A \vec{x}_{opt}) = 0$ for all $y \in \mathbb{R}^n$

for all vectors y in $\mathbb{R}^n$, orthogonal to them every other vector $\Rightarrow \vec{0}$

Hence, only $\vec{0}$ is orthogonal to all vectors in $\mathbb{R}^n$.

$$A^T b - A^T A x_{opt} = \vec{0}$$

$$A^T b = A^T A x_{opt}$$

$$\Rightarrow x_{opt} = (A^T A)^{-1} A^T b$$

PseudoInverse

$$\text{so, } A^+ = (A^T A)^{-1} A^T$$

In other means,

$$x_{opt} = \frac{A^T b}{A^T A} \quad (\text{like a projection formula}).$$

$A^T A \Rightarrow$ Gram Matrix in Gram-Schmidt is Invertible, for PseudoInverse

$$\text{so, } A A^+ = A (A^T A)^{-1} A^T$$

$$\text{so, } A x_{opt} = \text{Projection of } b = P b$$

$$A (A^T A)^{-1} A^T b = P b$$

$\Downarrow$

Projection of b

$\downarrow$

Idempotent and Symmetric matrix

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(Projection matrix)



$(A(A^T A)^{-1} A^T)^T \Rightarrow$ Verifying that this is
Symmetric

~~$(A(A^T A)^{-1} A^T)^T$~~

$(A^T)^T (A^T A)^{-1} A^T$

$A (A^T A)^{-1} A^T$

$\Rightarrow A (A^T A)^{-1} A^T$

$\Rightarrow$ Symmetric ✓

$A(A^T A)^{-1} A^T$ satisfied

not orthogonal (like a projection)
(column)

$\frac{A^T A}{A^T A} = I$

thin matrix is invertible $\Rightarrow A^T A$
invertible if columns are

$$A^T (A^T A)^{-1} A = I$$

That was about Tall Matrix,
now, wide matrix

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

matrix A

$$\text{Col space}(A) = \mathbb{R}^2$$

Because, Rank(A): Row Rank: 2

$$\Rightarrow \alpha_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ will span entire } \mathbb{R}^2$$

$$\therefore \begin{bmatrix} 2 \\ 3 \end{bmatrix} \text{ is there in } \mathbb{R}^2$$

$$A^T A \Rightarrow \text{Not Invertible}$$

$$A A^T \Rightarrow \text{Invertible (?)}$$

$$\text{that means, } (A A^T)(A A^T)^{-1} = I$$

here, $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is in $\mathbb{R}^2$, so, the projection of this into itself is

Identity matrix

$$\text{So, } A (A^T (A A^T)^{-1}) = I \Rightarrow \text{pb } \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix} \text{ (Doubt?)}$$

$$\Downarrow \boxed{A^T = A^T (A A^T)^{-1}} \text{ Pseudo Inverse.}$$

$A^{n \times n}$ (square matrix), rank = n

$$A^T = (A^T A)^{-1} A^T \text{ (we know)} = A^{-1} (A^T)^T$$

Now,

$$A A^T = A (A^T A)^{-1} A^T$$

$$\text{So, } \boxed{\text{Pseudo Inverse} = A^{-1}}$$



$A^\dagger$ exists, Pseudo Inverse $A^\dagger = A^+ = A^{-1}$

Suppose, A is a projection matrix

Then Pseudo Inverse of $A = A$ itself; $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$A^2 = A$$

$$A(A) = A$$

$$A^\dagger A = A \quad \text{if } A \text{ is projection matrix}$$

GSO (Gram Schmidt ~~Orthogonalization~~ Orthogonalization)

$$\vec{e} = \vec{u}$$

Generate $\vec{e}, \vec{f}, \vec{g}$ from $\vec{u}, \vec{v}, \vec{w}$

$$\vec{f} = \vec{v} - \left(\frac{\vec{v} \cdot \vec{e}}{\vec{e} \cdot \vec{e}} \right) \vec{e}$$

$$\vec{g} = \vec{w} - \left(\frac{\vec{w} \cdot \vec{e}}{\vec{e} \cdot \vec{e}} \right) \vec{e} - \left(\frac{\vec{w} \cdot \vec{f}}{\vec{f} \cdot \vec{f}} \right) \vec{f}$$

$$\vec{u} = \vec{e}$$

$$\vec{v} = \left(\frac{\vec{v} \cdot \vec{e}}{\vec{e} \cdot \vec{e}} \right) \vec{e} + \vec{f}$$

$$\vec{w} = \left(\frac{\vec{w} \cdot \vec{e}}{\vec{e} \cdot \vec{e}} \right) \vec{e} + \left(\frac{\vec{w} \cdot \vec{f}}{\vec{f} \cdot \vec{f}} \right) \vec{f} + \vec{g}$$



$$[e \ f \ g] \begin{bmatrix} 1 & (\vec{v} \cdot \vec{e})/(\vec{e} \cdot \vec{e}) & (\vec{w} \cdot \vec{e})/(\vec{e} \cdot \vec{e}) \\ 0 & 1 & (\vec{w} \cdot \vec{f})/(\vec{f} \cdot \vec{f}) \\ 0 & 0 & 1 \end{bmatrix} A$$

column vectors e, f, g as a matrix
and divide with their magnitudes.

$$= [a_1 \ a_2 \ a_3] \begin{bmatrix} 1/\|e\| & (\vec{v} \cdot \vec{e})/(\vec{e} \cdot \vec{e}) & (\vec{w} \cdot \vec{e})/(\vec{e} \cdot \vec{e}) \\ 0 & 1/\|f\| & (\vec{w} \cdot \vec{f})/(\vec{f} \cdot \vec{f}) \\ 0 & 0 & 1/\|g\| \end{bmatrix}$$

$A = QR$; QR factorization

$$A^{m \times n} = Q^{m \times n} R^{n \times n} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{1}{s} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

If $m \neq n$

$$(QR)^{-1} = R^{-1} Q^{-1} \quad \left(\text{how, } Q^{-1} = Q^T \right. \\ \left. \text{don't know} \right) \\ = R^{-1} Q^T \quad \text{should see}$$

$$\text{so } \boxed{A^T = R^{-1} Q^T}$$

If $n \times n$

$$(A^{n \times n})^{-1} = (QR)^{-1}$$

$$\Rightarrow \boxed{A^{-1} = R^{-1} Q^T}$$

$$AA^T = (QR)R^TQ^T$$

Ex: $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ u & v are given

$u = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$
 $v = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T$

$e = u = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$

$f = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$$= \begin{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$A = QR$ Factorization

$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} -1/2 \\ 1 \\ 1/2 \end{pmatrix}$

$\begin{bmatrix} 1 & -1/2 \\ 0 & 1 \\ 1 & 1/2 \end{bmatrix}$

$Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & \sqrt{2}/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$

$\sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{3}{2}}$

$-1/2 \times \sqrt{2}/\sqrt{3}$

The idea of Positive Definite, Negative Definite, etc will be needed when the time of Multivariate calculus starts,

Positive Definite RSM

All Eigen values are strictly positive

Negative Definite RSM

All Eigen values are strictly negative